

Ethan Shih

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EDUCATION

University of California, Irvine

Expected: 2029

Computer Science and Engineering, B.S.

Irvine, CA

- Relevant Coursework: Python Programming, C++ Programming, Digital Systems, Linear Circuit Analysis, Data Structures and Algorithms

EXPERIENCE

Software Subteam Member

2026 – Present

Legacy Robotics

- Developed a PyQt-based mission control GUI for a University Rover Challenge rover, with 4 mission-specific dashboard tabs each surfacing telemetry, rover status, sensor data, controls, and GNSS navigation, alongside a dedicated camera array window.
- Designed the GUI around interchangeable data sources, enabling simulation data for dashboard development and testing ahead of ROS2 integration.
- Collaborated with software and systems leads and a 25-person cross-disciplinary rover team to define dashboard requirements and plan ROS2 interfaces for telemetry, controls, and mission data.

Web Development Intern

May 2024 – August 2024

ClassChat

- Converted a Figma landing page design into a responsive, production-ready implementation using HTML, CSS, and JavaScript, building a custom image carousel and resolving absolute-positioned elements across device sizes and zoom levels.
- Collaborated remotely with designers and project leadership through iterative feedback and weekly progress updates to refine and deliver web features.

PROJECTS

NEXUS | *Python, FastAPI, SQLAlchemy, PostgreSQL, Next.js, React, TypeScript*

2026 – Present

- Built a full-stack volunteer logistics platform to replace the Google Forms and Sheets workflows for Science Olympiad directors, targeting deployment across multiple tournaments this year.
- Designed and implemented a multi-tenant FastAPI backend with 130+ REST endpoints across a 33-table PostgreSQL schema supporting tournament-specific, role-based access for volunteers and administrators.
- Developed a schema-aware form system that retains submitted responses after live form changes and implemented JWT and API-key authentication with scoped permissions.
- Wrote more than 1,200+ Pytest tests to support reliable feature development.
- Established contributor workflows using GitHub issues, feature branches, and pull requests for parallel development.

LEADERSHIP

Vice Captain and Cabinet Officer

2021 – 2025

Glen A. Wilson High School Science Olympiad

- Led team to 3rd place out of 150+ teams in SoCal – the school's highest-ever State placement – earning 100+ individual medals across Regionals, State, and Invitationals.
- Mentored middle school students in Helicopters and Towers events, leading to their first-ever State qualification.
- Directed STEM exploration camps and onboarded new members through hands-on training, fostering collaboration and long-term team development.

TECHNICAL SKILLS

Languages: Python, SQL, C++, TypeScript, Java, JavaScript, HTML/CSS

Frameworks: FastAPI, SQLAlchemy, Next.js, React, PyQt, Tailwind CSS, Node.js

Developer Tools: Git, GitHub, VS Code, SQLite3, PostgreSQL